

Classifying Land Cover for Microclimate Analysis with Machine Learning

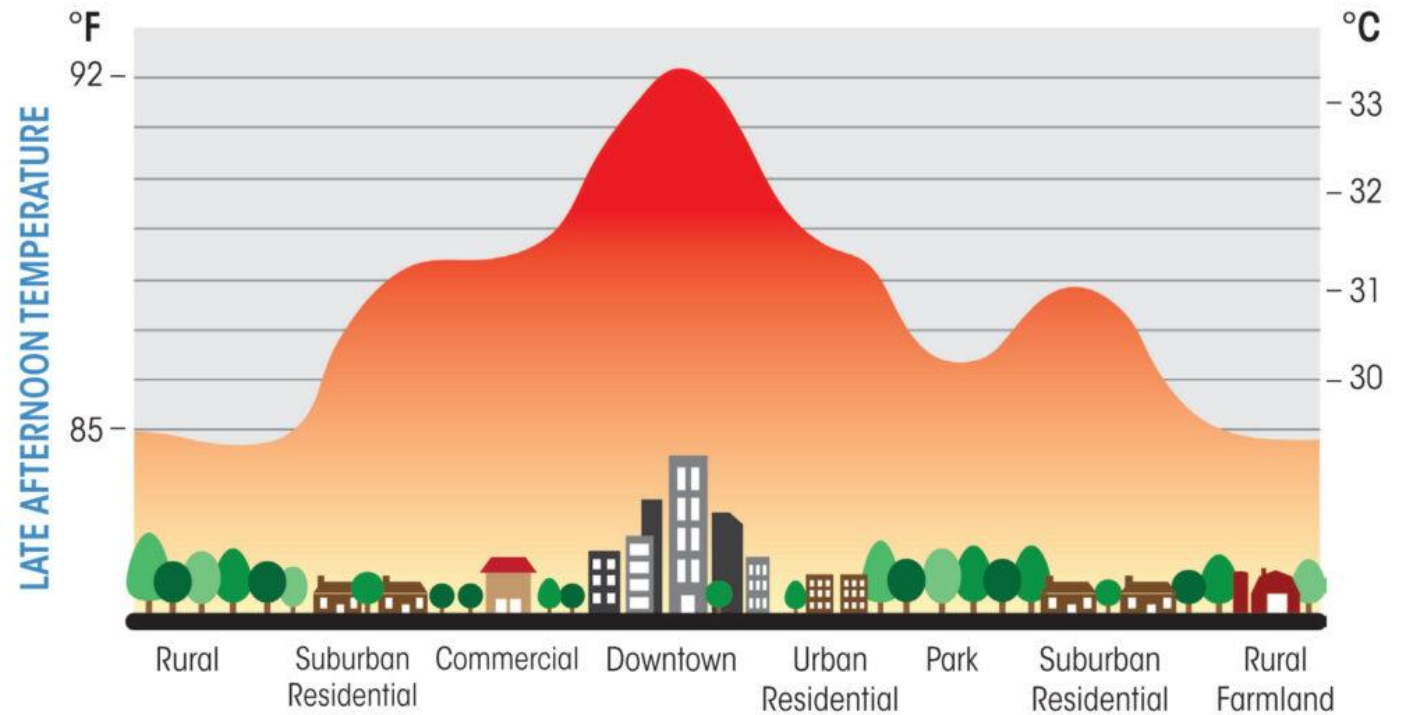
February 17, 2026

Introduction & Project Goal

Context: Microclimates are driven by land cover (urban, vegetation, water) and play key role in urban planning.

The Problem: Manual land cover mapping is slow.

The Goal: To build an automated, accurate classifier using imagery to support environmental analysis.



Source: Google images

Data & Methodology Overview

Experience (Data): 300 aerial images of an agricultural region in California's Central Valley collected by University of California Merced.

- **Features:** Color Histograms (color) + HOG (texture/shape).
- **Final Feature Vector:** 1860 dimensions per image.

Task: Classify land cover (Buildings, Forest, River).

Performance: Metrics: Accuracy (primary) + Macro F1-Score (balanced across classes).

Models compared: Multinomial Logistic Regression (baseline), Perceptron, SVM with RBF kernel.

- **Validation Strategy:**
 - **70/15/15 Train/Dev/Test split** (test set kept completely unseen)
 - **5-Fold Cross-Validation** on training set.
 - **GridSearchCV** for hyperparameter tuning (SVM: C, gamma)

Methodology Overview...

Computations

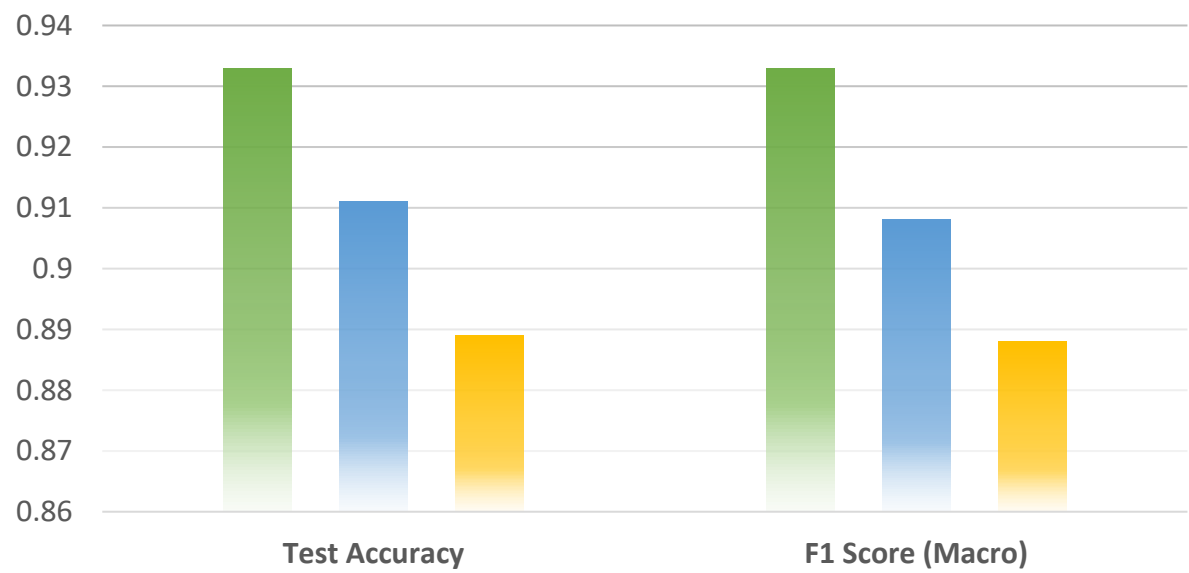
The code is written in Python programming language and compute following operations:

- Loads 3-class images and extracts HOG + Color Histogram features.
- Creates Train/Dev/Test splits (70%/15%/15%) and scales features.
- Trains three models: Logistic Regression, Perceptron, SVM (with CV tuning).
- Final evaluation on the untouched Test set
- Generates diagnostic plots: Confusion Matrices & Learning Curve.

Results

Performance: Model Comparison on Test Set

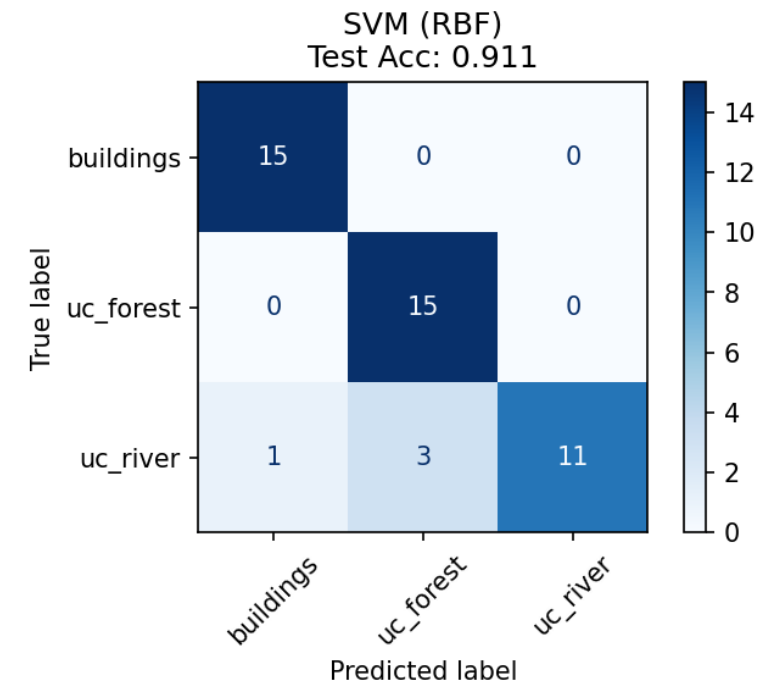
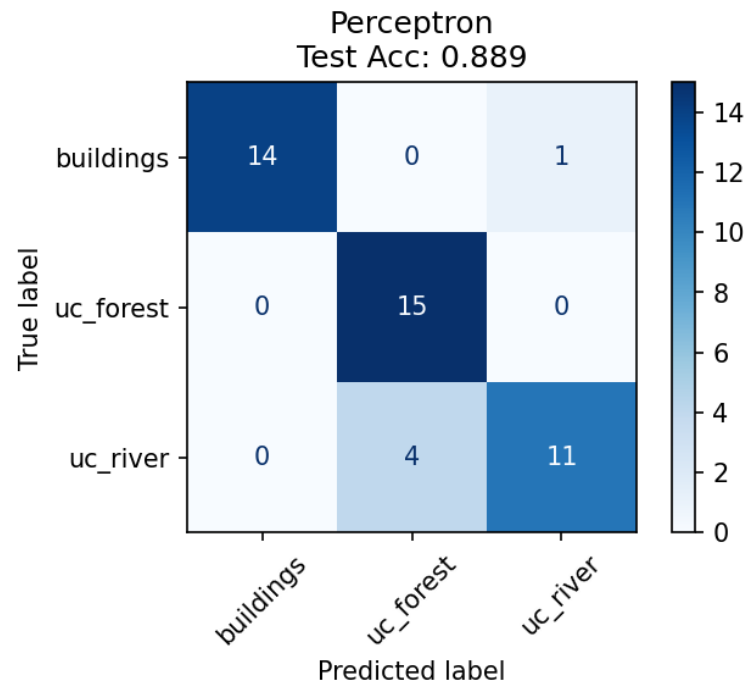
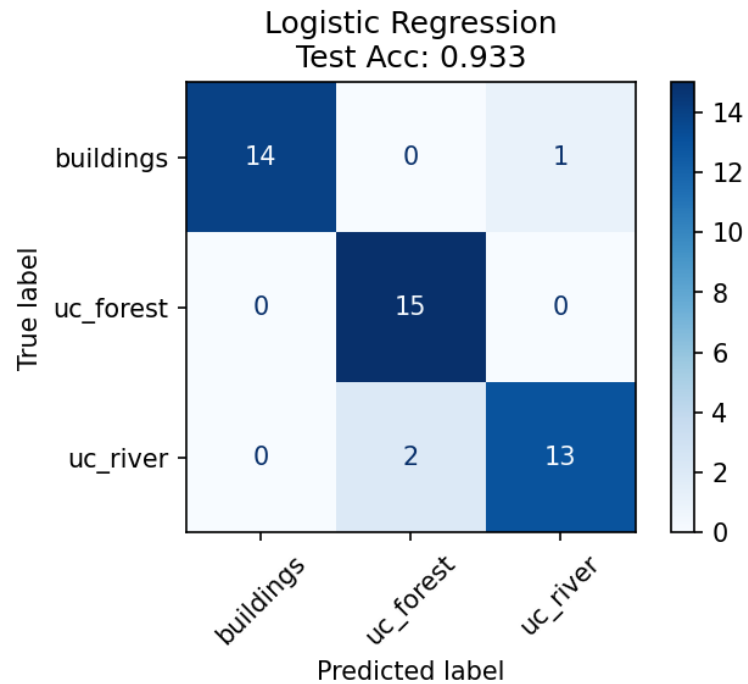
Model	Test Accuracy	Test F1-Score (Macro)
Logistic Regression	0.933	0.933
SVM with RBF Kernel	0.911	0.908
Perceptron	0.889	0.888



Results...

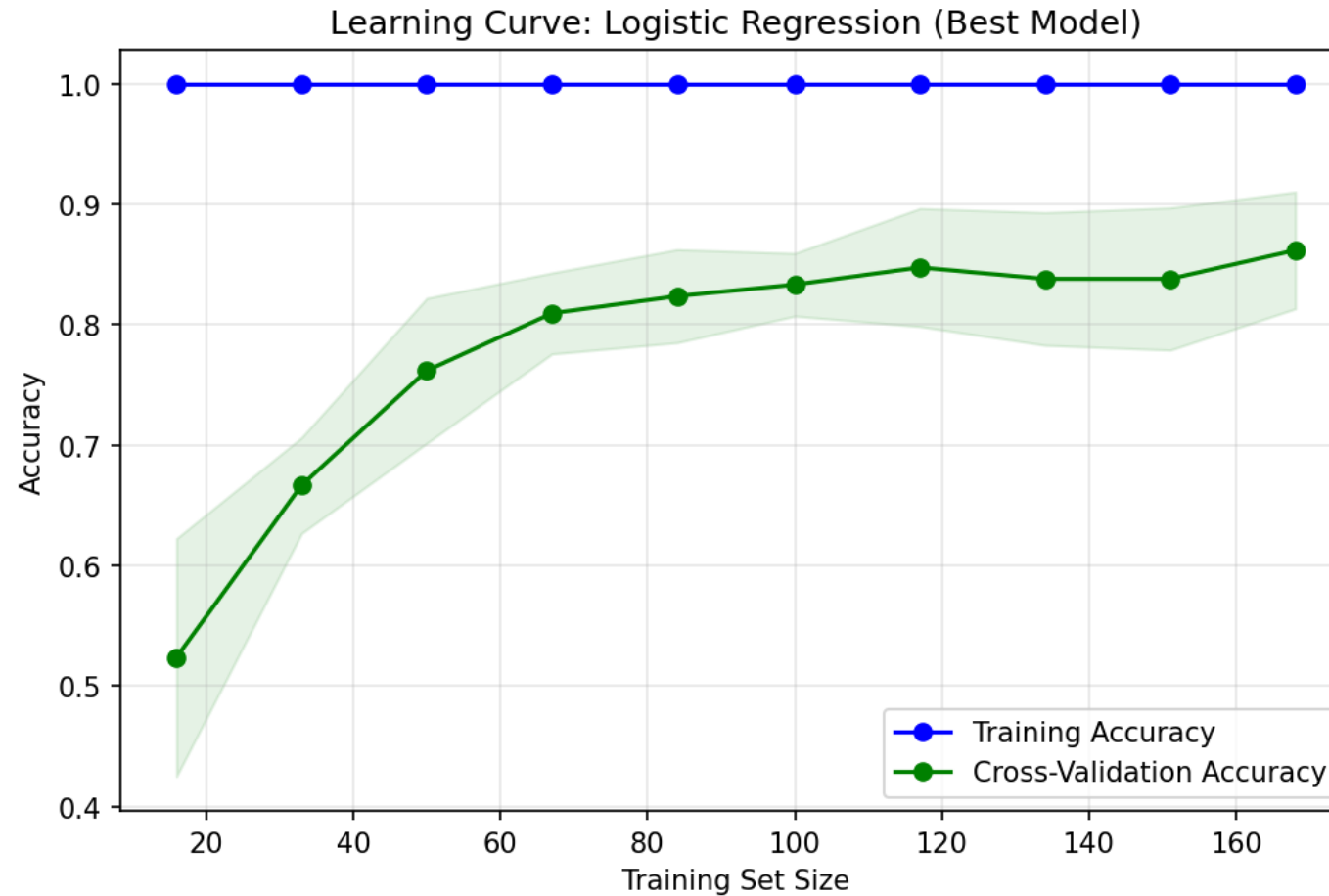
Where Models Confuse (The Diagnostic)

Confusion Matrices - Test Set Performance



Results...

Performance: Diagnosing Bias vs. Variance (Learning Curve for Best Model)



Conclusion

- Features (HOG + Color) were highly effective for this task, making advanced models like SVM unnecessary.
- The learning curve shows uniform convergence – as training set size increases, empirical risk converges to true risk.
- The hypothesis class (linear decision boundaries) provides an ϵ -representation of the true concept in our feature space.
- With 1860 features and 210 training samples, our hypothesis class can shatter the data points, though the actual VC dimension of logistic regression ensures we still generalize well.

Successfully built a (PAC learner) classification model with >93% accuracy using traditional ML.



Thank you for your attention!